

Quantum Double Models (QDM)

Kitaev, Annals of phys., 303, 2 (2003)

arXiv 1997

QDM: generalization of $\mathbb{Z}_2$ toric code
to lattice G gauge theory.

→ finite (may be non-Abelian) group.

$$D(G) = G \times \hat{G} \text{ for Abelian } G.$$

$\downarrow$ $\downarrow$
 flux charge

2.1. The model based on a group algebra.

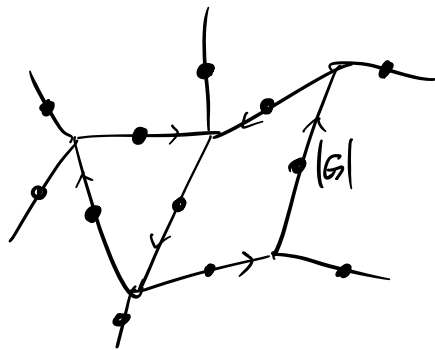
G .

$\mathcal{H} := \mathbb{C}[G] = \left\{ \sum_g a_g \cdot g \mid g \in G, a_g \in \mathbb{C} \right\}$ be the group algebra.
 $\downarrow$
 Hilbert space with orthonormal basis $|g\rangle$
 $\dim \mathcal{H} = |G|$

$$\begin{aligned} \text{Def: } L_+^g |z\rangle &= |gz\rangle & T_+^h |z\rangle &= \delta_{h,z} |z\rangle \\ L_-^g |z\rangle &= |zg^{-1}\rangle & T_-^h |z\rangle &= \delta_{h^{-1},z} |z\rangle \end{aligned}$$

$$\begin{aligned} \text{Satisfy: } L_+^g T_+^h &= T_+^{gh} L_+^g, & L_+^g L_-^h &= T_-^{hg^{-1}} L_+^g \\ L_-^g T_+^h &= T_+^{hg^{-1}} L_-^g, & L_-^g T_-^h &= T_-^{gh} L_-^g \end{aligned}$$

Hilbert space:



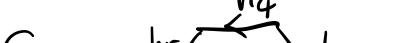
$|G|$ -dim vector space on
each link of a 2D lattice.

$$\overrightarrow{\bullet} = \overleftarrow{\bullet}, \quad g \in G.$$

QDM : $H = - \sum_s A_s - \sum_p B_p$

$$\bigvee A_s^g := \prod_{\substack{\text{edge } l \\ \text{of } s}} L_{\pm}^g, \quad A_s := \frac{1}{|G|} \sum_{g \in G} A_s^g$$

↳ gauge transf.

B_p

 $= \underbrace{\delta_{h_1 h_2 \dots h_6, h}}_{\text{flux is } h}$


$$B_p^h = \sum_{h_1 \dots h_6 = h} \prod_{m=1}^6 T_{\pm}^{h_m}, \quad B_p := B_p^{h=1}$$

↪ zero flux condition for p.

relation $\left\{ \begin{array}{l} A_s^2 = A_s \\ B_p^2 = B_p \\ [A_s, A_{s'}] = 0 \\ [B_p, B_{p'}] = 0 \\ [A_s, B_p] = 0 \end{array} \right\}$ projectors

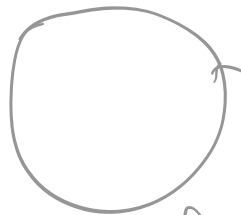
A_5

$$\begin{array}{ccc} g_2 \downarrow & \xrightarrow{\phi} & g_2 g_1 \\ s & \xrightarrow{g_1} & \end{array} = \begin{array}{ccc} g_2 g_1^{-1} \downarrow & \xrightarrow{\phi} & (g_2 g_1^{-1}) \cdot (g_1 g_3) = g_2 g_1 \\ & & \xrightarrow{g_3} \end{array}$$

Ground state : $A_s |\psi\rangle = B_p |\psi\rangle = |\psi\rangle$, $\forall s, p$
↓ gauge transf. ↓ flat connection condition

$\Rightarrow |\psi\rangle$: flat connections up to gauge equivalence.

T.C. $|\psi\rangle = | \bigcirc \rangle + | \bigodot \rangle + \dots$
 $g = e^{iA} \in U(1)$



flat : $\oint_C A_\mu dx^\mu = 0$
↓

$\prod g_1 \dots g_N = 1 \Leftrightarrow B_p^{h=1}$

$g_{ij} = e^{iA_{ij}}$ ↗ continuous
↘ discrete

$g = e^{iA}$
 $g^{-1} = e^{-iA}$
 r
 $g = e^{iA}$
 $g = e^{-iA}$

Gauss law

$\nabla \cdot \vec{A} = \rho$

$\parallel$

$A(\vec{r} + \hat{x}) - A(\vec{r})$

$+ A(\vec{r} + \hat{y}) - A(\vec{r})$

$\downarrow$

$g_{\vec{r}+\hat{x}} g_{\vec{r}}^{-1} g_{\vec{r}+\hat{y}} g_{\vec{r}}^{-1}$

Important notice:

The courses of week 3 (on 2021-09-27 and 2021-09-29) will be cancelled. And the ending date of the course will be postponed to week 13 of the fall semester.

From 2021-10-04 on, we will use another Tencent Meeting Room [9952830954](#) (password 654321) of the university account, such that there are more storage space for the recordings.

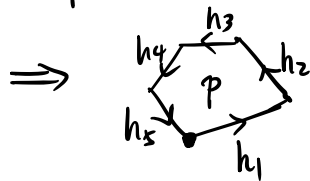
Recall. QDM.

$$H = - \sum_s A_s - \sum_p B_p$$

$$[A_s, B_p] = 0$$

$$\textcircled{1} B_p |\psi\rangle = |\psi\rangle, \quad \forall p$$

$$\xrightarrow{h} = \xleftarrow{h^{-1}}$$



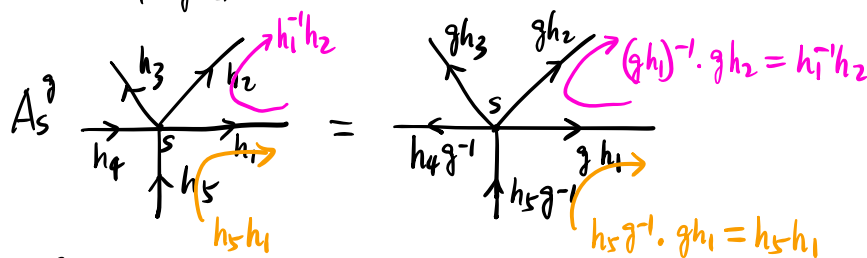
$$h_1 h_2 \cdots h_5 = 1 \in G.$$

(flat connection condition)
(zero flux condition)

$$\textcircled{2} A_s |\psi\rangle = |\psi\rangle, \quad \forall s$$

$$A_s := \frac{1}{|G|} \sum_{g \in G} A_s^g$$

A_s^g : gauge transformation.



A_s^g fluctuate $\{h_{ij}\}$ within the subspace $B_p=1, \forall p$.

$\Rightarrow G_s |\psi\rangle$ is an equal weight superposition of all flat connections that are gauge equivalent.

continuous:

U(1) gauge theory

$$A_\mu(x) \rightarrow A'_\mu = A_\mu - \partial_\mu \chi$$

$$\oint_C A'_\mu dx^\mu = \oint_C (A_\mu - \partial_\mu \chi) dx^\mu = \oint_C A_\mu dx^\mu = \oint$$

is gauge inv.

discrete:



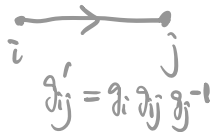
$$g_{ij} = e^{i \int_{site i}^{site j} A_\mu dx^\mu}$$

→ gauge field

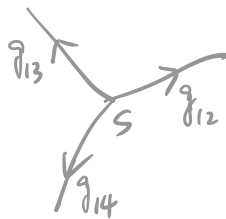
$$g_i = e^{i \chi(r_i)}$$

→ gauge transf. para.

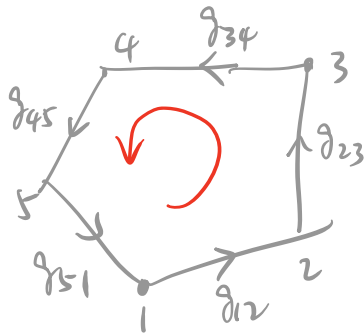
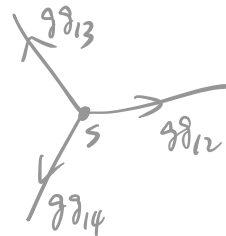
$$\begin{aligned} g'_{ij} &= e^{i \int_i^j (A_\mu - \partial_\mu \chi) dx^\mu} = e^{i \int_i^j A_\mu dx^\mu - \chi(r_j) + \chi(r_i)} \\ &= e^{i \chi(r_i)} e^{i \int_i^j A_\mu dx^\mu} e^{-i \chi(r_j)} \\ &= g_i \cdot g_{ij} \cdot g_j^{-1} \end{aligned}$$



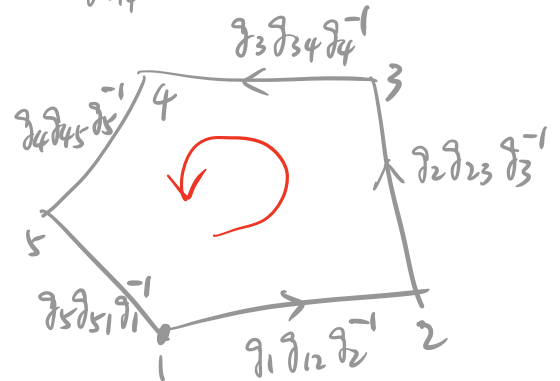
eg:



$A_s^g \rightarrow$



$\{A_s^g\}$
gauge transf.



$$\begin{aligned} \oint &= g_{12} g_{23} \dots g_{51} \\ &= e^{i \int_1^1 A_\mu dx^\mu} \end{aligned}$$

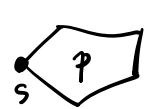
$$\begin{aligned} \oint' &= g_1 g_{12} g_2^{-1} \cdot g_2 g_{23} g_3^{-1} \cdot \dots \\ &\quad \cdot g_5 g_{51} g_1^{-1} \\ &= g_1 (g_{12} \dots g_{51}) g_1^{-1} \\ &= g_1 \oint g_1^{-1} \end{aligned}$$

$$\text{Tr } \oint' = \text{Tr } \oint$$

2.2. Anyon excitations and ribbon operators

$$H = - \sum_s \underbrace{\frac{1}{|G|} \sum_{g \in G} A_s^g}_{\text{proj. to gauge inv. subspace}} - \sum_p \underbrace{B_p^{h=1}}_{\text{proj. to } h=1 \text{ zero flux}}$$

Excitations: ① non gauge inv.: $\frac{1}{|G|} \sum_g A_s^g \neq 1$
 ② non zero flux: $B_p^{h \neq 1}$

To understand the excitations of QDM, we first try to understand the algebra of A_s^g, B_p^h for 

$$\begin{cases} A^g A^{g'} = A^{gg'} \\ (A^g)^+ = A^{g^{-1}} \\ B^h B^{h'} = \delta_{hh'} B^h \\ (B^h)^+ = B^h \\ A^g B^h = B^{ghg^{-1}} A^g \end{cases}$$

Def $D_{(h,g)} = h \bigcup_g := B_h \cdot A_g$

$$D_{(h_1, g_1)} \cdot D_{(h_2, g_2)} = \delta_{h_1, g_1 h_2 g_1^{-1}} D_{(h_1, g_1 h_2)}$$

The algebra generated by $D_{(h,g)}$ is called the Drinfeld quantum double $D(G)$ of G .

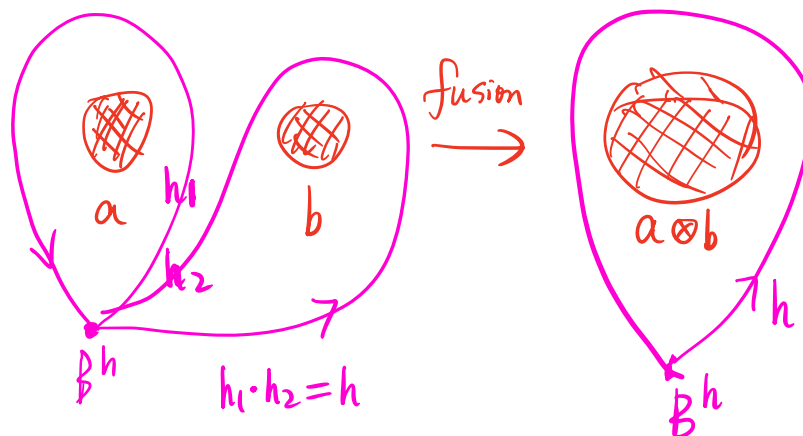
① Co-algebra structure $\Delta: D(G) \rightarrow D(G) \times D(G)$ → how to act on two anyons by one $D(G)$ element

$\Delta(A^g) = A^g \otimes A^g$ → act on anyon a

$\Delta(B^h) = \sum_{h_1 h_2 = h} B^{h_1} \otimes B^{h_2}$

$\Rightarrow \text{Rep } D(G)$ is a tensor/monoidal category.

$$a \otimes b \rightarrow c$$



② $D(G)$ is quasi-triangular: $R = \sum_{g \in G} B^g \otimes A^g$

braiding

$\Rightarrow \text{Rep } D(G)$ is a braided tensor category.

Excitation space supports a representation of $D(G)$.

Irreps of quantum double $D(G)$
 $\longleftrightarrow$ anyon excitation types
 (braiding, fusion)

Irreps of $D(G)$.

Let $u \in G$, $C = \{gug^{-1} \mid g \in G\}$ the conjugacy class,

$E = \{g \in G \mid gu = ug\}$ the centralizer.

Claim. There is one irreducible rep (C, χ) for each conjugacy class C and each irrep χ of the centralizer group E .

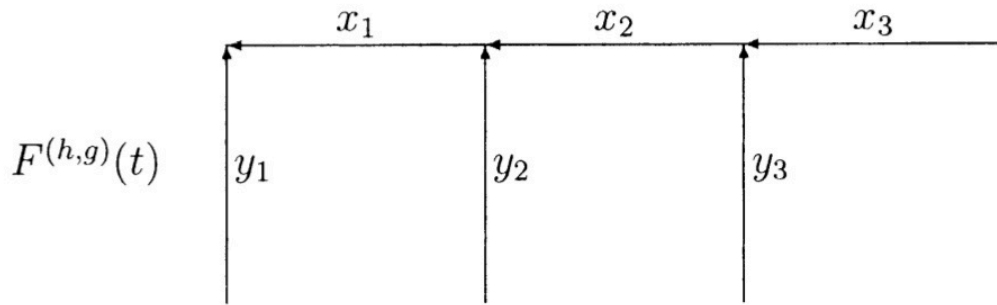
flux
 $\uparrow$
 change

Ribbon operators



Create a pair of anyonic excitations at the ends of the ribbon operators.

↳ finite width (framing)

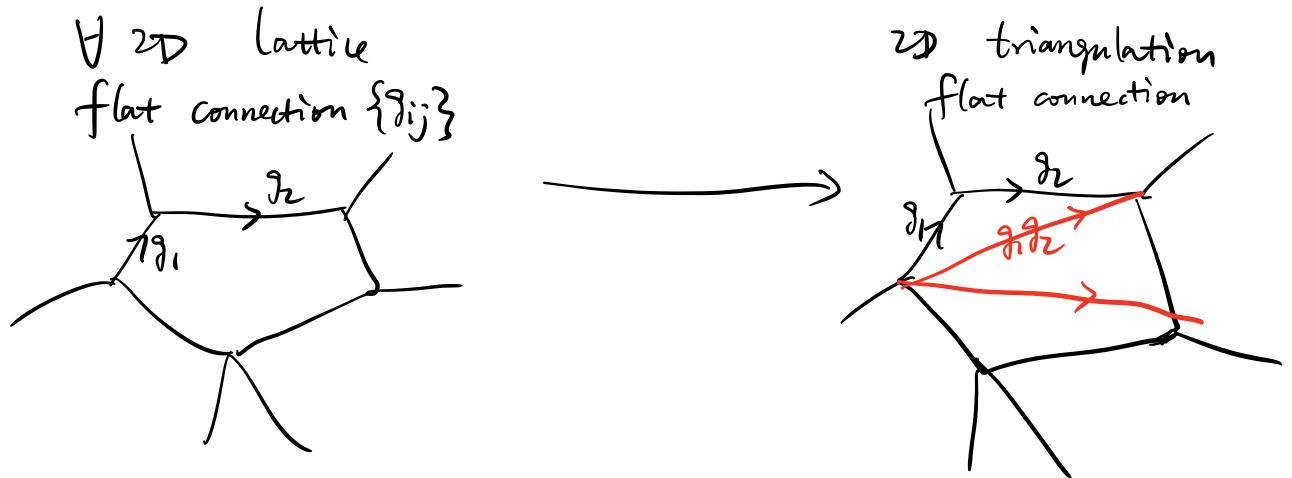


$$= \delta_{g, x_1 x_2 x_3} \begin{array}{c} \begin{array}{ccc} x_1 & x_2 & x_3 \\ \hline h y_1 & (x_1^{-1} h x_1) y_2 & ((x_1 x_2)^{-1} h (x_1 x_2)) y_3 \end{array} \end{array}$$

$$\begin{aligned} [F^{(h,g)}(t), B_p] &= 0 \\ [F^{(h,g)}(t), A_s] &= 0 \end{aligned} \quad \text{for } s, p \neq \partial t$$

F gives us fusion, braiding of anyons.

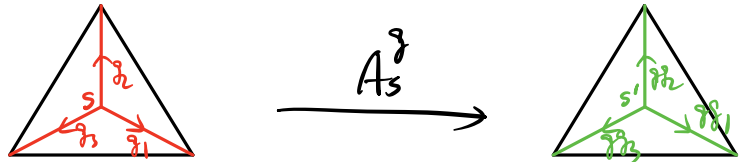
2.3. Spacetime Path Integral Picture



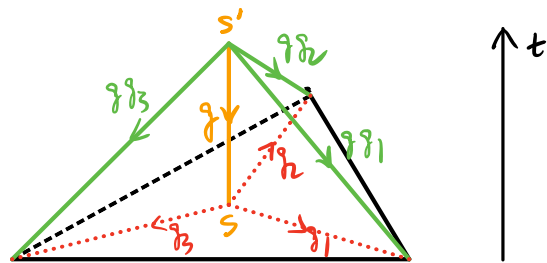
B_p operator: defined for $\forall$ triangle.

A_s^g operator:

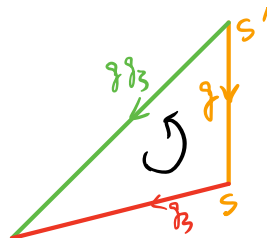
Space :



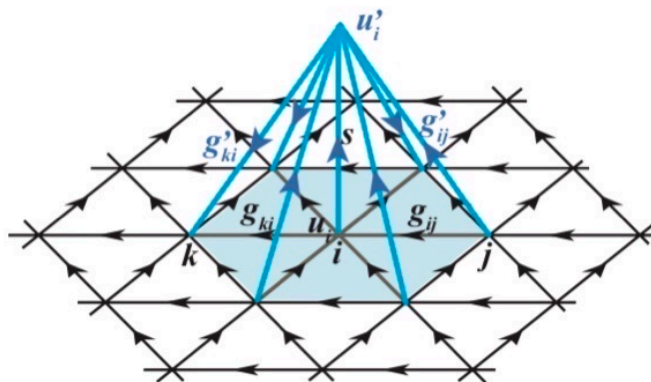
spacetime:

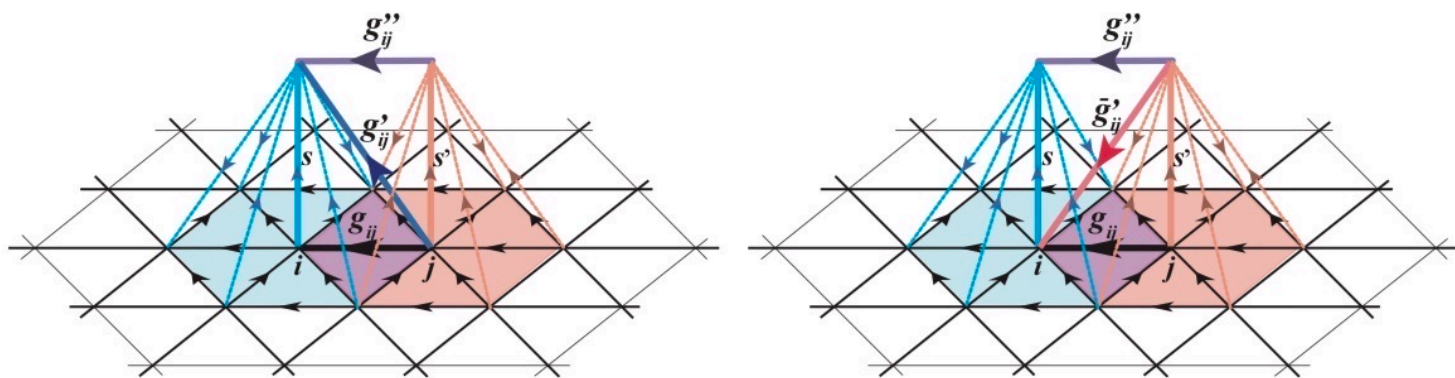


new link along t direction



flat connection for new spacetime triangles.





$$A_j^{\partial'} A_i^{\partial} = A_i^{\partial} A_j^{\partial'}$$